

SIMATS ENGINEERING
ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME:

Cloud Computing and
Big Data Analytics

COURSE CODE:

CSA1520

COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

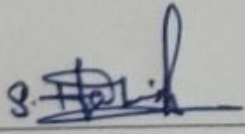
Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 4

Total Marks: 20

Code of Conduct

I S. Harish/192511119 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

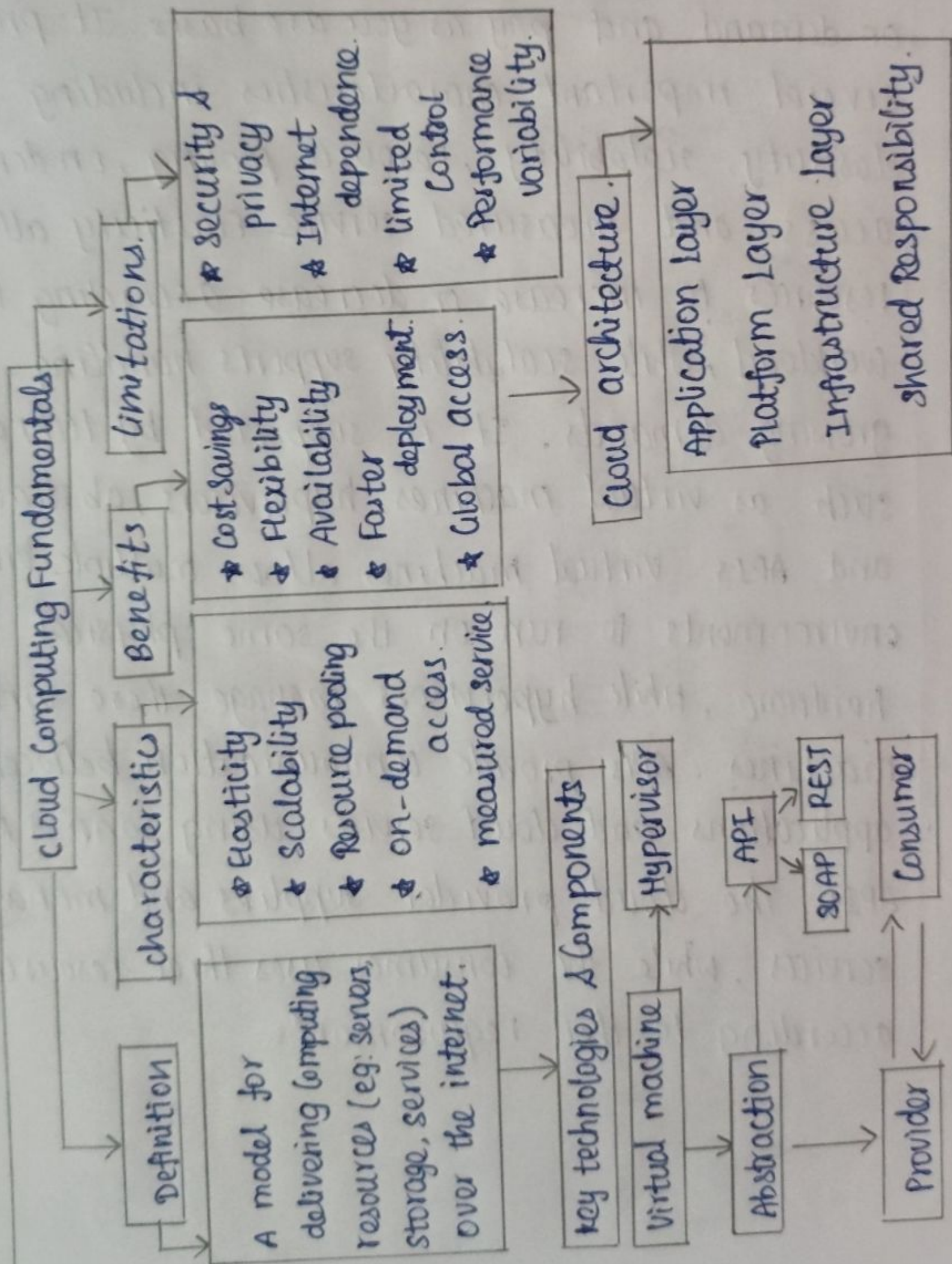
1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

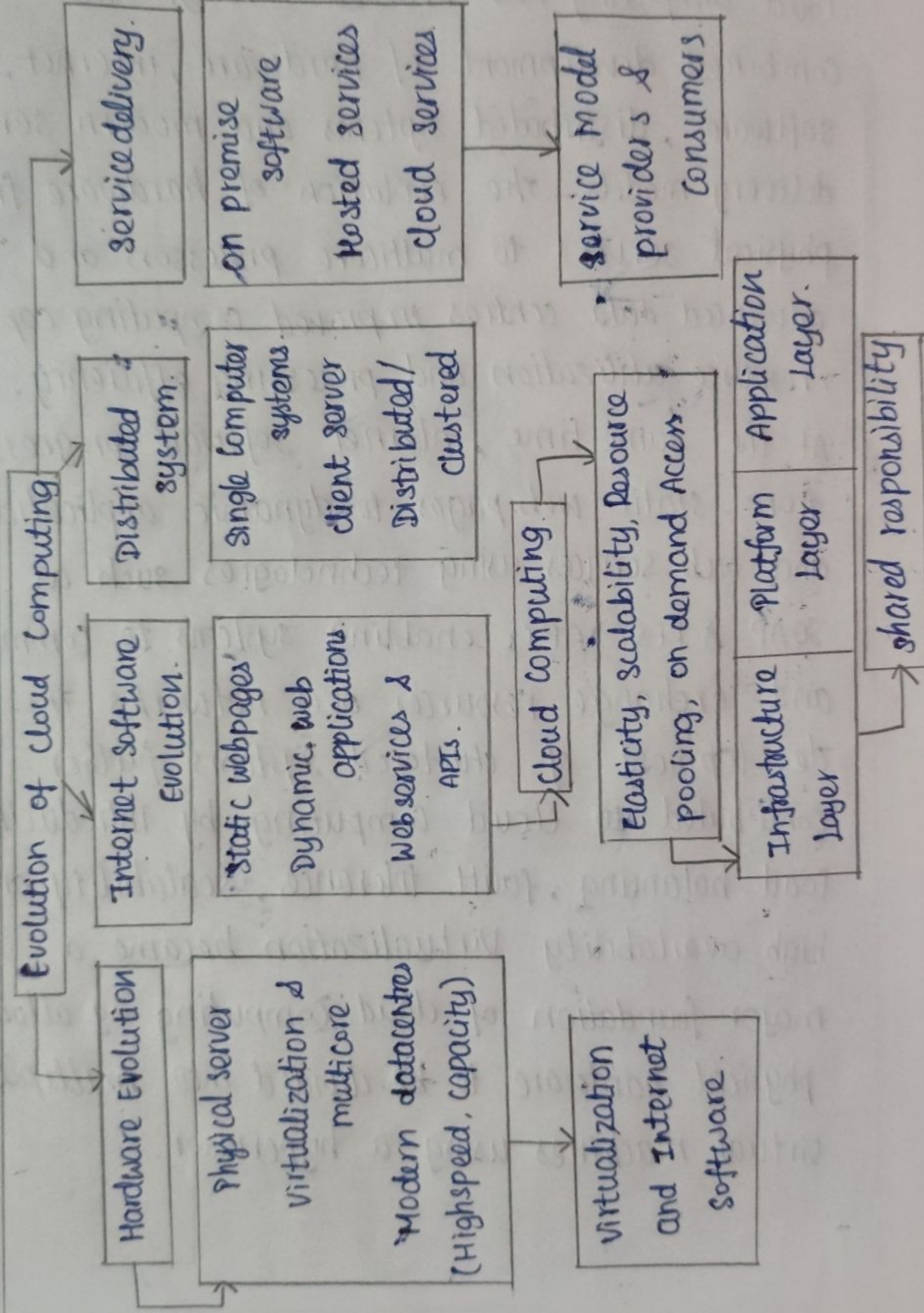
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship Mapping	25%	Links between concepts are accurate and meaningful	Most links are correct	Some links are weak or unclear	Relationships are mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented



cloud computing is a model for delivering computing resources such as servers, storage, applications and networking services over the internet on an on demand and pay as you use basis. It provides several important characteristics including elasticity, scalability, resource pooling, on demand access and measured service. Elasticity allows resources to increase or decrease according to workload, while scalability supports handling growing demands. It is supported by technology such as virtual machines, hypervisors, abstraction and APIs. Virtual machines allows multiple virtual environments to run on the same physical hardware, while hypervisors manage these virtual machines. APIs provide communication between applications and cloud services using SOAP & REST APIs. The cloud provider supplies and manages services, while the consumer uses these resources according to their requirements.

2)



cloud computing has evolved through the combined development of hardware, internet, software, distributed systems and modern service delivery models. The evolution of hardware from physical servers to multicore processors and advanced data centres improved computing capacity, resource utilization and processing efficiency. At the same time, internet software progressed from static web pages to dynamic applications and web services using technologies such as SOAP & REST APIs, enabling systems to communicate and exchange resources over networks. The development of clustered systems further contributed to cloud computing by introducing load balancing, fault tolerance, scalability and high availability. Virtualization became a major foundation of cloud computing by allowing physical hardware to be divided into multiple virtual machines using a hypervisor.

Cloud Service models.

Infrastructure as a service (IaaS).

virtualized computing resources. Physical infrastructure, virtualization layer, hypervisor, security & availability, network & usage.

Platform as a service (PaaS)

development and runtime platform. for build, testing & deploying. Infrastructure, virtualization, Middleware.

software as a service (SaaS).

Ready to use software applications accessible over the internet. Platform, Data storage.

Anything as a service (XaaS)

extended cloud services (eg: SaaS, FaaS, BaaS, Infra, platform, service management, APIs, security & monitoring).

key characteristics of cloud services

Elasticity, scalability, resource pooling, ondemand Access and measured service.

Provider

↕ via APIs

Consumer

cloud layers

Infrastructure Layer

Platform Layer

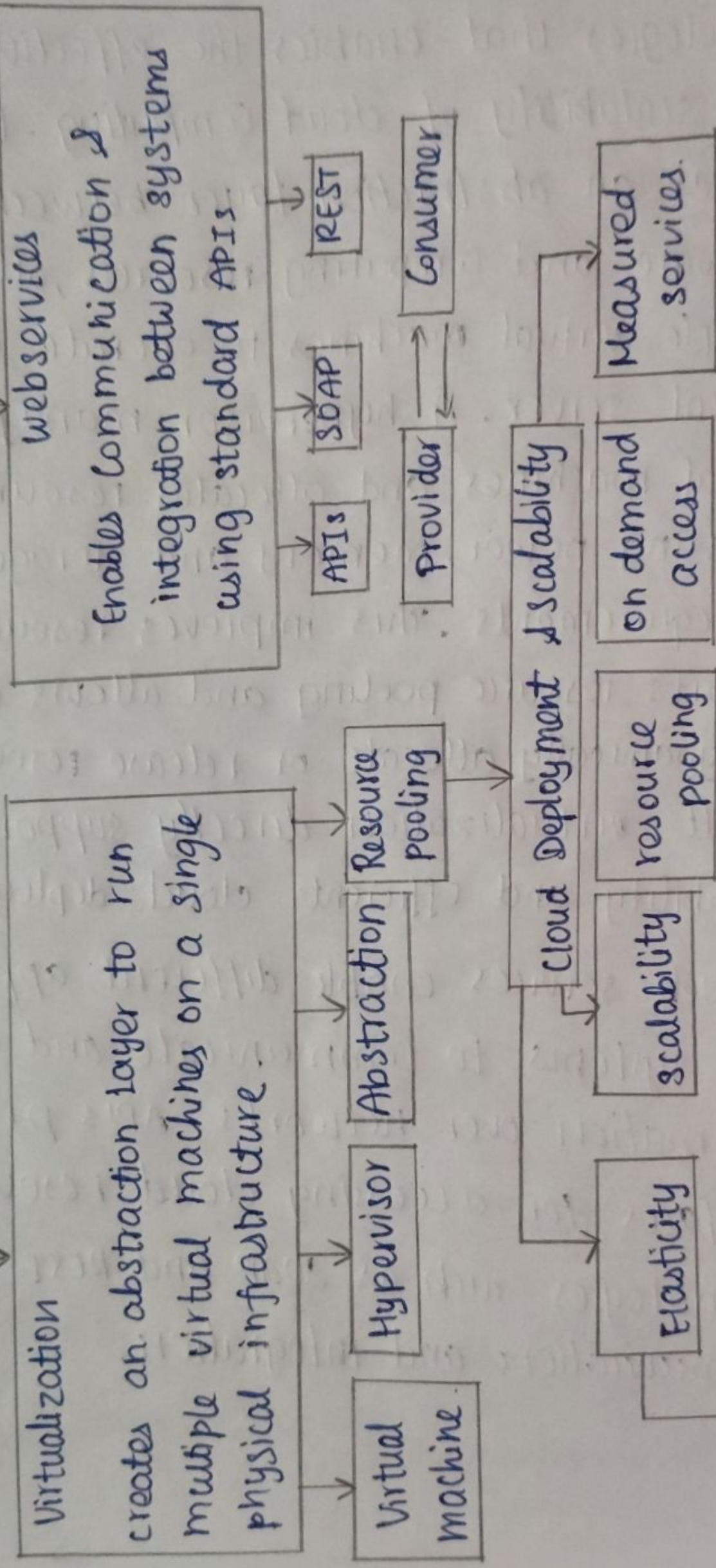
Application Layer.

shared responsibility

cloud service models define how computing resources and services are provided to consumers through cloud platforms. The three primary models are Infrastructure as service (IaaS), Platform as a Service (PaaS), Software as a service (SaaS) while Anything as a service (XaaS) represents the broader range of services delivered through the cloud. IaaS provide virtualized infrastructure such as virtual machines, storage and networking with the provider managing the physical infrastructure, virtualization and hypervisor while the consumer manages the operating system, applications, data & configurations.

PaaS provides a complete platform for developing, testing and deploying applications. The provider manages the infrastructure, runtime environment, middleware and development tools, while the consumer mainly manages application code, data and settings. SaaS provides ready to use software applications over the internet.

Virtualization and Web services
as enablers of cloud.



virtualization and web services are two important technologies that enable the effective deployment and scalability of cloud computing. Virtualization creates an abstraction layer between physical hardware and computing resources, allowing multiple virtual machines to operate on a single physical server. A hypervisor manages these virtual machines and allocates resources such as processing power, memory and storage according to requirements. This improves resource utilization, supports resource pooling and allows cloud providers to dynamically allocate or release resources. As a result, virtualization directly supports elasticity, scalability and efficient cloud deployment.

web services enable different applications and systems to communicate and exchange information over networks. APIs provide standard interfaces for accessing cloud resources, while technologies such as SOAP and REST support service communication and integration.